Syllabus for MATH 218, Fall 2026, Applied Linear Algebra (3 credit hours)

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DISCLAIMER

Course Structure:

Math 218 lectures meet on campus, in person, Mondays, Wednesdays, and Fridays. There are no discussion sections. Individual instructors may record their classes if they choose to, but you are not entitled to a recording if you miss class.

Instructors:

- Mercer “Tabes” Bridges mbridg5@uic.edu
- Daniel Groves dgroves@uic.edu
- Olga Kashcheyeva kolga@uic.edu
- John Steenbergen (Coordinator) jbergen@uic.edu
- Gregory Taylor gtaylo9@uic.edu

Class meeting times and rooms:

- Dr. Taylor’s 8am section: 8:00-8:50am MWF in LCD D004
- Dr. Groves’ 9am section: 9:00-9:50am MWF in LCB B101
- Dr. Groves’ 10am section: 10:00-10:50am MWF in LCD D004
- Dr. Steenbergen’s 12pm section: 12:00-12:50pm MWF in SES 130
- Dr. Kashcheyeva’s 2pm section: 2:00-2:50pm MWF in LCD D002
- Dr. Bridges’ 4pm section: 4:00-4:50pm MWF in LCF F001

No matter which section you are in, please take note: *the two midterm exams in Math 218 are held in the evening, **not** at the regular class time!*

Instructor Drop-in Office Hours:

- Dr. Bridges: M 1:00-1:50pm (MSLC) and W 1:00-1:50pm (SEO 309)
- Dr. Groves: W 1:00-1:50pm (SEO 727) and F 1:00-1:50pm (MSLC)
- Dr. Kashcheyeva: M 4:00-4:50pm (SEO 718) and W 1:00-1:50pm (MSLC)
- Dr. Steenbergen (coordinator): MWF 11:10-11:50am (MSLC)
- Dr. Taylor: W 10:00-10:50am (SEO 1217) and F 10:00-10:50am (MSLC)

Note: there may be more office hours added later when our TA graders have been assigned. All updated office hours will be listed on Canvas.

UIC Email

We care about your success. When you have concerns to discuss or accommodations you need, please contact your instructor at your earliest convenience.

Email will be the main form of communication. Please, check your UIC email a couple of times a day so you do not miss out on any important announcements. Email announcements should also appear on our Canvas course page in case you don't have access to email. You are required to use your UIC email address for all aspects of this course.

In emails, always identify yourself and what class and section you are in. Please, put the course number (Math 218) and class time (for example, MWF 2pm) in the subject line. This helps your instructor identify course related emails. Please, be professional and courteous in your email.

If you email us and do not hear back from us within two business days, then please send us a gentle reminder.

Course Materials

A first course in linear algebra, version **2023-B**, by Kenneth Kuttler and Lyryx Learning, is our official course textbook. This book is available for free online: a PDF that can be downloaded is available on Canvas.

We will use MyOpenMath for online homework, which is also free. Students will access it via Canvas and should not have to make an account.

Students seeking an additional resource are encouraged to consult *Linear algebra and its applications*, **fifth** edition, by David Lay, Steven Lay, and Judi McDonald. This is the book that was formerly used in Math 218 at UIC and contains many good problems. A hardcopy of the book is not required but can be purchased online from many websites.

Course Description

According to the UIC course catalog, Math 218 covers the following topics: "matrices, Gaussian elimination, vector spaces, *LU* decomposition, orthogonality, the Gram-Schmidt process, determinants, inner products, eigenvalue problems, diagonalization of symmetric matrices, and applications to differential equations and

Markov processes." We will cover most of Chapters 1–4 and 7, and the first half of Chapter 5, of the Kuttler/Lyryx textbook.

Prerequisites and credits

A grade of C or better in Math 181 (Calculus II) is required in order to remain enrolled in Math 218. Credit is *not* given for both Math 218 and Math 310 (which is no longer offered). Math 218 is a 3-credit-hour course.

Learning Objectives

Upon successful completion of Math 218, students will be able to translate between systems of linear equations and associated (augmented) matrices, use matrix algebra to solve linear systems, use determinants to test for invertibility of square matrices, understand the concepts of span, linear independence/dependence, subspace, and basis, use the Gram-Schmidt process to find an orthogonal basis for a given subspace, find the closest vector in a subspace to a given vector, find the eigenvalues and eigenvectors of a square matrix (a linear transformation from $\mathbb{R}^n$ to itself) and use this information to determine large powers of the matrix and "long-run" behavior of systems encoded by the matrix, and use the singular value decomposition of a matrix to approximate it by lower-rank (more "compressible") matrices.

Course Copyright

We retain copyright of all materials developed for this course. Please DO NOT upload any component of our course materials onto third-party websites (this includes uploading screenshots of the questions or typing up the questions from MyOpenMath homework, written homework, exam, and/or practice problems in Chegg). Please do not share our course materials with anyone who is not enrolled in this course.

Grading and Assessments

The course grade will be distributed as follows:

- Online Homework: 15%
- Written Homework: 15%
- Midterm Exam #1: 20%

- Midterm Exam #2: 20%
- Final Exam: 30%

Midterm Exam #1 will be 6:00-7:30pm on Monday, Oct. 5th, and Exam #2 will be 6:00-7:30pm on Thursday, Nov. 12th. The final exam will be during finals week (May. 4-8th), exact date and time TBD.

The grading scale is:

- 87%-100%: A
- 75%-86.99%: B
- 60%-74.99%: C
- 45%-59.99%: D
- 0%-44.99%: F

We guarantee an A to students with a combined final course percentage of 87% or higher, a B to students with a percentage of 75% or higher, a C to students with a percentage of 60% or higher, and a D to students with a percentage of 45% or higher. The department reserves the right to be more lenient than this, but there is no curve and no other way to improve your grade other than what is mentioned in this syllabus.

There is no required minimum score on any of the exams in order to pass the course; all students with a final weighted average above 60% will pass.

Online Homework

This semester we are using a free service, MyOpenMath, for online homework. Students will be able to access MyOpenMath (“MOM”) through the course Canvas page. You are allowed to use calculators or other technology (such as WolframAlpha) when completing homework.

There will be 6 online MyOpenMath homework assignments total. Each assignment will become available about two weeks before it’s due. Most MyOpenMath homework problems allow several attempts per question as well as the option to replace a question with a similar one (some multiple-choice questions are exceptions to this, allowing only one or two attempts; for problems of this type, your score will drop with each subsequent wrong answer).

Written Homework

We will have 6 written homework assignments over the course of the semester, which students will submit using Gradescope. A link to Gradescope is available on Canvas as well as instructions for how to submit assignments to it.

Each written homework assignment will be posted to both Gradescope and Canvas about two weeks before it's due. The problem sets will be written and graded in a similar way to exam problems. You must write your responses by hand if possible.

Use of Technology on Assignments

Math 218 is a course on applied linear algebra. Real-world applications of linear algebra often involve large matrices, and much of the tedious work (such as row reduction or matrix inversion) is "outsourced" to technology. This course is not going to require you to use technology, but we are going to allow and even encourage you to use it *where appropriate*.

WolframAlpha (https://www.wolframalpha.com), specifically, is free and can be easily used by anyone with an Internet browser. In particular, on online homework (but not exams or written homework), you will be allowed to use technology, such as a calculator or a program like WolframAlpha. Our focus is going to be on intelligent application and interpretation of the results of matrix computations, which is much harder to outsource to a machine.

Exams and Exam Dates

Exams will be held on campus, in person. There will be no online exams. If you need accommodations for any aspect of this course then please contact your instructor and the course coordinator immediately.

Please mark your calendars for these dates and time:

Exam #1 6:00 PM - 7:30 PM, Monday, October 5th

Exam #2 6:00 PM - 7:30 PM, Thursday, November 12th

Final Exam TBD (sometime during finals week, December 7th-11th)

Room locations will be announced about a week before each exam. Be sure to bring your ID card and writing utensils for exams. You will not be allowed any other items

besides those—no notes, books, calculators (or other devices), etc. Being found with AI glasses or with any AI client open on any device during an exam will result in a score of 0 for that exam.

There will be 5 extra points awarded on the final exam for all students with a good attendance record. See our attendance policy below.

Attendance

Attendance in all lectures is very strongly encouraged to maximize your learning and engagement. We have seen improvements in student performance for those that attend class regularly. Your instructor will be taking attendance most likely using Acadly, a free online app you can download onto your phone. When using Acadly, be sure to register using your UIC login (not a private email address).

Attendance is not strictly required and does not directly go into your final score, but anyone with a good attendance record will be awarded 5 extra points on the final exam. Specifically, of the last 36 lectures in which attendance is taken, if you have missed no more than 10 lectures then you will have 5 extra points added to your final exam score at the end of the course.

Make-up Policy

Your lowest online homework score, and your lowest written homework score will each be dropped. That is, your overall score for each will be based on your five best scores in that category.

Additionally, in MyOpenMath, each student will be given two “latepasses”. A latepass can be used to extend the deadline of an online homework assignment by 48 hours. Use of the two latepasses is at the student’s discretion.

Further extensions beyond the two latepasses, as well as extensions to the written homework assignments will not generally be granted (but it never hurts to ask). There will be no make-up exams except in cases where a student has a prior approval from their instructor due to a conflict with the scheduled exam times.

Weekly Schedule

Week	Topics	Homework/Exams
Week 1 8/24-8/28	Systems of linear equations, coefficient and augmented matrices, REF and RREF, row operations	
Week 2 8/31-9/4	Solution sets, homogeneous systems, rank	
Week 3 9/7-9/11	Applications of linear systems, matrix operations, matrix inverses No class on 9/7: Labor Day	MOM HW #1 due Tuesday, 9/8
Week 4 9/14-9/18	More on matrix inverses, linear transformations	Wr. HW #1 due Tuesday, 9/15
Week 5 9/21-9/25	More on linear transformations, *LU* factorization	MOM HW #2 due Tuesday, 9/22
Week 6 9/28-10/2	Definition of the determinant, properties of the determinant	Wr. HW #2 due Tuesday, 9/29
Week 7 10/5-10/9	Span, linear independence, subspaces, basis, dimension	MOM HW #3 due Tuesday, 10/6 Midterm Exam #1 Monday, 10/5, 6-7:30pm, Location TBA
Week 8 10/12-10/16	Bases for subspaces, null and column spaces of matrices, the rank-nullity theorem	Wr. HW #3 due Tuesday, 10/13
Week 9 10/19-10/23	Eigenvalues and eigenvectors, the characteristic polynomial	MOM HW #4 due Tuesday, 10/20
Week 10 10/26-10/30	Diagonalization, large powers of diagonalizable matrices, Markov chains	Wr. HW #4 due Tuesday, 10/27
Week 11 11/2-11/6	Dot products, orthogonal projection, orthogonal sets and orthogonal matrices, the Gram-Schmidt process	MOM HW #5 due Tuesday, 11/3
Week 12 11/9-11/13	QR factorization, orthogonal projection onto a subspace, least-squares approx.	Wr. HW #5 due Tuesday, 11/10 Exam #2 Thursday, 11/12, 6-7:30pm, Location TBA
Week 13	The spectral theorem, orthogonal	MOM HW #6 due Tuesday, 11/17

Week	Topics	Homework/Exams
11/16-11/20	diagonalization, singular values, SVD	
Week 14 11/23-11/27	Applications of SVD, the matrix exponential No classes 11/25 - 11/27: Thanksgiving	
Week 15 11/30-12/4	Review for final exam	Wr. HW #6 due Tuesday, 12/1
Finals Week 12/7-12/11	No classes this week	Final Exam date/time TBD

Course Conduct

We are committed to creating a learning environment where diverse perspectives are recognized and valued as a source of strength. We request that all students work with us to create a class culture based on open communication, mutual respect, and inclusion. As a class we will approach all discussions with respect and civility. Disagreements and debates in academic discourse are expected and welcome, but personal attacks are never OK, and will not be tolerated. We strive to ensure an open and welcoming classroom for all students. If we ever miss the mark, please don't hesitate to come and talk to us. We are all learning together.

Please do not carry on conversations with friends, or work in groups on homework or projects for other courses, while your instructor is presenting course material. Students who disrupt the learning environment for others in this way may be asked to leave class.

Please do not use a chatbot to generate emails to your instructor. If we receive emails that were clearly composed by a chatbot, we reserve the right to delete them without responding.

Academic Integrity Policy

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community - students, staff, faculty, and administrators - share the responsibility of ensuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the Student Disciplinary Policy. Academic dishonesty on an exam will result in a score of 0 for that exam and may incur harsher penalties in certain cases. Academic dishonesty on a homework assignment will result in a score of 0 on that assignment. Multiple instances of academic dishonesty on homework assignments will incur harsher penalties.

Examples of academic dishonesty include but are not limited to:

- Submitting work that is copied from another student
- Sharing full solutions to homework assignments or exams with other students. (You are encouraged to work together with other students on homework assignments, but please do not share full solutions.)
- Using ChatGPT or other generative AI to do your homework or exams for you
- Using any sort of outside help on exams other than your own brain

You are capable of meeting expectations for this course. If you are concerned about how well you are doing in this course, please come speak with your instructor instead of considering academic misconduct.

Use of Generative Artificial Intelligence

For online homework assignments, the use of AI writing tools (including, but not limited to, ChatGPT, Bard, or Sudowrite) can be a powerful tool to help you learn **if used responsibly**. *Remember, the goal is to help you understand how to do the problems, not just do the problems for you*. However, the use of AI writing tools is NOT permitted for written homework or exams in this course. Students who use these tools irresponsibly for class assignments undermine the goals and learning objectives for this course, reducing the effectiveness of instruction. The instructor may submit student writing to an AI writing detector (e.g., GPTZero) at any point throughout the term. Any confirmed use of AI writing tools will be treated as cheating. Students should reference UIC's Student Disciplinary Policy for more information.

Alternatives to Academic Dishonesty

- **Seek out help** – meet with your TA or Professor, ask if there is special tutoring or other arrangements available.
- **Drop the course** – can you take it next semester when you might feel more prepared and less pressured?
- **See a counselor** at Student Psychological Services, and/or your school, college or department advisor– UIC has many resources for students who are feeling the stresses of academic and personal pressures, see resources below.

Remember, **getting caught cheating affects more than just your GPA**. How will you explain to your parents, family and friends that you have been suspended or dismissed? How will it affect your financial aid award and/or scholarship money? How will it affect your future career plans?

You have worked very hard to get here, so don't cheat! You don't need to. If you would like more information, please see the Dean of Students' Office in 3030 Student Services Building (SSB), 1200 W. Harrison Street, call at (312) 996-4857, or visit their website at https://dos.uic.edu/.

Student Wellness Resources

As an undergraduate at UIC, you may experience challenges such as struggles with academics, finances, student life, or your personal well-being. Please know this is completely normal and that you shouldn't hesitate to ask for help. Come to us, or if it is about an issue beyond the scope of this class, please contact your college advisors, or get help from any number of other support services and resources available to all UIC students:

- Current Student Resources (a one-stop shop for links to resources in the following categories: General, Academic, Student Support, Student Life, Technology, Health and Safety, and Getting Around Campus)
- UIC Tutoring Resources
- Reserve library space to study, take an exam, or participate in online classes while on east campus(Daley Library) or west campus (Library of the Health Sciences).
- Access U&I Care Program for assistance with personal hardships.
- Financial Assistance for the course materials can be requested on the Dean of Students website through the U & I Care Fund.

Importantly, if you are in immediate distress, please call the UIC Counseling Center at (312) 996-3490 to speak directly with a counselor or to schedule an appointment with a counselor. If calling after business hours (which at UIC are typically from 9 am -5 pm, M-F), press 2 to be connected to a crisis counselor. You can find additional online mental health resources on the Counseling Center Resources page.

Syllabus Statement

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. Students who face or anticipate disability-related barriers while at UIC should connect with the Disability Resource Center (DRC) at drc.uic.edu, drc@uic.edu, or at (312) 413-2183 to create a plan for reasonable accommodations. In order to receive accommodations, students must disclose disability to the DRC, complete an interactive registration process with the DRC, and provide their course instructor with a Letter of Accommodation (LOA). Course instructors in receipt of an LOA will work with the student and the DRC to implement approved accommodations.

Academic Deadlines

Current academic calendar and the list of deadlines can be found at the university catalog website.

STUDENT RESOURCES FOR SUPPORT

Counseling Services

Counseling services are available for all UIC students. You may seek free and confidential services from the Counseling Center counseling.uic.edu. The Counseling Center is located in the Student Services Building; you may contact them at (312) 996-3490. In addition to offering counseling services, the Counseling Center also operates the InTouch Crisis Hotline from 6:00 p.m.-10:30 p.m. They offer support and referrals to callers, as well as telephone crisis interventions; please call (312) 996-5535.

Campus Advocacy Network

Under the Title IX law you have the right to an education that is free from any form of gender- based violence and discrimination. Crimes of sexual assault, domestic violence, sexual harassment, and stalking are against the law and can be prevented. For more

information or for confidential victim-services and advocacy contact UIC's Campus Advocacy Network at 312-413- 1025 or visit http://can.uic.edu/. To make a report to UIC's Title IX office, email TitleIX@uic.edu or (312) 996-5657.

Academic Accommodations based on a Disability

The University of Illinois at Chicago is committed to full inclusion and participation of people with disabilities in all aspects of university life. Students who face or anticipate disability-related barriers while at UIC should connect with the Disability Resource Center (DRC). You may contact DRC at 312-413-2183 (v) or 312-413-0123 (TTY) and consult the following website .

In order to receive accommodations, students must disclose disability to the DRC, complete an interactive registration process with the DRC, and provide their course instructor with a Letter of Accommodation (LOA). Course instructors in receipt of an LOA will work with the student and the DRC to implement approved accommodations.

Religious Holidays

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when he/she will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, he/she may request remedy through the campus grievance procedure.

Grievance Procedures

UIC is committed to the most fundamental principles of academic freedom, equality of opportunity, and human dignity involving students and employees. Freedom from discrimination is a foundation for all decision making at UIC. Students are encouraged to study the University's "Nondiscrimination Statement". Students are also urged to read the document "Public Formal Grievance Procedures". Information on these policies and procedures is available on the University web pages of the Office of Access and Equity.

School-Life Conflict

Many students face obstacles to their education because of work or family obligations or unforeseen personal difficulties. If you are experiencing challenges throughout the term that are impacting your ability to succeed in this course, or in your undergraduate career more broadly, please reach out to your instructor immediately so that we can work together to form a plan for your academic success.

DISCLAIMER

The instructors reserve the right to make changes to the above policies as the course progresses if circumstances change or such an original policy is not in line with student learning.